

## Ideation Phase

### Define the Problem Statements

|                            |                                                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b>       | India's Agricultural Crop Production Analysis                                                                                                                                                   |
| <b>Domain</b>              | Agriculture / Data Analytics & Business Intelligence                                                                                                                                            |
| <b>Tool</b>                | Tableau Desktop / Tableau Public                                                                                                                                                                |
| <b>Dataset</b>             | Historical Indian Crop Production Data, 1997–2021                                                                                                                                               |
| <b>Prepared By</b>         | Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)                                                                  |
| <b>Mentor</b>              | N/A — Individual Submission                                                                                                                                                                     |
| <b>GitHub Repository</b>   | <a href="https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis">https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis</a>             |
| <b>Tableau Public Link</b> | <a href="https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject">https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject</a> |
| <b>Date</b>                | 02/07/2026                                                                                                                                                                                      |

### Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

#### Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

#### Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

#### Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

### Customer Problem Statement — Two Primary Personas

| Problem Statement (PS)         | I am (Customer)                                                                          | I'm trying to...                                                                                      | But...                                                                                                 | Because...                                                                                                | Which makes me feel...                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| PS-1 (Rajesh — Farmer)         | A small-scale farmer in Uttar Pradesh with limited access to agricultural data           | Decide which crops to grow each season to maximize yield and income                                   | I don't have access to historical crop performance data for my region or season                        | Agricultural data is complex, scattered across government reports, and not presented visually             | Anxious, uncertain about my income, and forced to rely on guesswork and tradition             |
| PS-2 (Ananya — Policy Analyst) | A government policy analyst at the Ministry of Agriculture responsible for policy design | Identify regions with low crop yield and understand long-term production trends to allocate resources | Data is fragmented across departments and not visualized in a way that supports quick policy decisions | Government datasets lack interactive visualization tools that allow drill-down by state, season, and year | Frustrated and unable to confidently justify resource allocation decisions with data evidence |

### Key Takeaway

Both personas are blocked by the same root cause: agricultural data exists but is not accessible or visual. Rajesh cannot see multi-year trends for his own district, and Ananya cannot compare regions quickly enough to act on them. This shared root cause justified a single Tableau solution serving both audiences (and, by extension, Vikram the investor) rather than three separate tools — one shared data model with role-specific dashboards layered on top.